

// COMPLETE REFERENCE GUIDE - STR1X3V0.XYZ

Windows 10 & 11 PC Optimization From Scratch

A complete, no-fluff guide to setting up and optimizing Windows for gaming and performance. Written for anyone — whether you just installed Windows for the first time or want to make sure you have not missed anything.

19

CHAPTERS

100+

TWEAKS

Both

WIN 10 & 11

MARCH 2026

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STR1X3V0.XYZ

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FREE TO SHARE

BEFORE YOU START

This guide covers both Windows 10 and Windows 11. Where steps differ between the two, both versions are explained. Commands shown in code blocks should be run in **PowerShell** — some require it opened as **Administrator** (right-click Start → Windows PowerShell (Admin) or Terminal (Admin)).

WHO THIS IS FOR

Anyone who has done a fresh Windows install, bought a new PC, or wants to audit their existing setup. No prior knowledge assumed. Every step explains what it does and why — not just how.

WHAT THIS DOES NOT COVER

- CPU or GPU overclocking — hardware-specific, covered elsewhere
- Undervolting — requires per-chip testing
- Third-party optimizer software — most are useless or harmful

// GOLDEN RULE

Do not install any "PC optimizer", "registry cleaner", or "game booster" software. Everything in this guide is done manually through Windows itself.

DIFFICULTY TAGS

TAG	MEANING
EASY	GUI clicks — no technical knowledge needed
MODERATE	PowerShell commands or registry edits — follow exactly
ADVANCED	BIOS or low-level settings — read warnings carefully

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First Boot & Privacy Settings

Configure Windows correctly from the moment it starts — before installing anything.

DURING THE SETUP WIZARD

ACCOUNT TYPE

Use a Local Account

When prompted to sign in with a Microsoft account, choose **Offline account** (Win10) or **Continue with limited setup** (Win11). A local account keeps your data on your machine. You can always add a Microsoft account later if needed for the Store.

PRIVACY SCREEN

Disable All Optional Data Collection

Turn all toggles off. Location, diagnostic data, tailored experiences, advertising ID. None improve your experience or performance.

OPTIONAL FEATURES

Skip OneDrive, Cortana, and Teams

Skip or decline all of these unless you actively use them. They add startup items and background processes.

AFTER SETUP – PRIVACY SETTINGS

Go to **Settings → Privacy** (Win10) or **Settings → Privacy & Security** (Win11):

SECTION	WHAT TO CHANGE
General	Disable advertising ID, website language access, app launch tracking, suggested content
Diagnostics & feedback	Set to Basic / Required only. Disable tailored experiences.
Activity history	Uncheck both boxes, click Clear
Location	Disable if not needed

Background apps
(Win10)

Disable for apps you do not need running in background

DISABLE CORTANA (WINDOWS 10)

Cortana runs in the background using CPU and network resources. Disable via **Settings → Cortana → Talk to Cortana → Off**. Then right-click taskbar → Cortana → Hidden.

DISABLE TIPS AND NOTIFICATIONS

Settings → System → Notifications & actions → disable **Get tips, tricks and suggestions** and **Show me the Windows welcome experience**.

Windows Update & Drivers

Get the OS and all hardware fully updated before installing anything else.

STEP 1: RUN WINDOWS UPDATE TO COMPLETION

Before installing any software or drivers, go to **Settings** → **Windows Update** → **Check for updates**. Install all updates and reboot. Repeat until nothing is left.

// WHY THIS MATTERS

Installing GPU or chipset drivers before Windows is fully updated can cause conflicts. Always update Windows first.

STEP 2: INSTALL DRIVERS IN THE CORRECT ORDER

FIRST

Chipset Drivers

Tell Windows how to communicate with your motherboard, CPU, and connected devices. Without them, other devices may not initialize at full capability.

AMD CPU: amd.com/support **Intel CPU:** intel.com/support

SECOND

GPU Drivers – Always Clean Install

A clean install removes all previous driver files before installing fresh. Dirty installs can leave corrupted files causing crashes and stutters.

NVIDIA: nvidia.com/drivers → Custom → tick **Perform a clean installation**

AMD: amd.com/support → select **Factory Reset** during install

THIRD

Audio Drivers

Get from your **motherboard manufacturer support page** — not from realtek.com. These are customized for your specific board.

FOURTH

LAN / WiFi Drivers

Windows usually installs these automatically but the generic versions may lack features. Get from your motherboard support page.

LAST

Peripheral Software

Mouse, keyboard, headset, and other peripheral software after everything else.

// IMPORTANT

Always download drivers from official manufacturer websites only. Third-party "driver updater" sites often bundle malware or serve outdated versions.

PAUSE AUTOMATIC UPDATES

Stop Windows force-restarting during a session: **Settings → Windows Update → Pause updates** for up to 5 weeks. Check manually on a regular schedule instead.

BIOS: XMP / EXPO & Key Settings

The most overlooked optimization — your RAM is almost certainly running slower than it should be.

XMP AND EXPO: RUN YOUR RAM AT ITS RATED SPEED

RAM sold as 3200MHz, 3600MHz, or 6000MHz does **not** run at that speed by default. Out of the box it runs at a universal safe speed — usually **2133MHz** or **2400MHz**. The full rated speed only activates when you enable the profile in BIOS.

PLATFORM	PROFILE NAME	WHERE TO FIND IT
Intel	XMP (Extreme Memory Profile)	AI Tweaker, Extreme Tweaker, or OC section
AMD	EXPO or DOCP	OC, DRAM, or Memory section

HOW TO ENABLE

Enabling XMP / EXPO

1. Restart and press **Delete** or **F2** to enter BIOS (varies by board)
2. Find XMP, EXPO, or DOCP — usually in Advanced Mode (F7 on most ASUS boards)
3. Set to **Profile 1**
4. Save and exit — press **F10** and confirm

VERIFY IT WORKED

After booting, open **Task Manager** → **Performance** → **Memory**. The speed shown should match your RAM packaging.

// IF WINDOWS DOES NOT BOOT AFTER ENABLING XMP

Reset the BIOS to defaults by removing the coin cell CMOS battery for 30 seconds. Your PC will boot with safe defaults and you can try again.

OTHER USEFUL BIOS SETTINGS

Resizable BAR (ReBAR)

Allows the CPU to access full GPU memory at once instead of in small chunks. Can improve performance in many games. Requires RTX 3000+ or RX 6000+ and a compatible motherboard. Look for **Resizable BAR** or **Above 4G Decoding**. Also ensure **CSM is disabled** (requires UEFI mode).

Boot Order

Set your Windows drive as the first boot device to avoid the BIOS checking other devices on every boot.

Power Plan

Stop Windows throttling your CPU to save power while gaming.

WHY THE DEFAULT PLAN IS WRONG FOR GAMING

The **Balanced** plan reduces CPU clock speeds when it detects low activity, then ramps back up when it detects high activity. This ramp-up introduces latency that appears as micro-stutters and inconsistent frame times in games.

OPTION 1: HIGH PERFORMANCE

Control Panel → Hardware and Sound → Power Options → High Performance. If not visible, click Show additional plans.

OPTION 2: ULTIMATE PERFORMANCE (RECOMMENDED)

Disables CPU core parking entirely. Must be unlocked — open admin PowerShell:

```
powercfg -duplicatescheme e9a42b02-d5df-448d-aa00-03f14749eb61
```

Then go to Control Panel → Power Options and select **Ultimate Performance**.

// LAPTOP USERS

High and Ultimate Performance drain battery faster. Use Balanced on battery, High/Ultimate when plugged in.

PROCESSOR POWER MANAGEMENT

In Power Options → Change plan settings → Change advanced power settings → Processor power management: set both **Minimum processor state** and **Maximum processor state** to 100%.

Startup Programs

Every unnecessary auto-start program eats memory and CPU before you have launched anything.

HOW TO AUDIT STARTUP PROGRAMS

Task Manager (Easiest)

Ctrl + Shift + Esc → Startup tab. Right-click any entry → Disable.

PowerShell (Full List)

```
Get-CimInstance Win32_StartupCommand | Select-Object Name, Command, Location | Format-Table -AutoSize
```

WHAT TO KEEP VS DISABLE

ALWAYS KEEP	KEEP IF USED CONSTANTLY	DISABLE – OPEN MANUALLY
Windows Security / Defender	Discord	Spotify
Audio drivers	Mouse / keyboard software	Steam / Epic / EA
GPU driver tray	Cloud storage (if syncing)	NVIDIA Broadcast
		RGB lighting software
		Torrent clients, Teams, OneDrive

DISABLE VIA POWERSHELL

```
# User-level entries
Remove-ItemProperty -Path "HKCU:\Software\Microsoft\Windows\CurrentVersion\Run" -Name "AppNameHere"

# System-level entries (admin PowerShell)
Remove-ItemProperty -Path "HKLM:\Software\Microsoft\Windows\CurrentVersion\Run" -Name "AppNameHere"
```

// RULE OF THUMB

Game Mode & HAGS

Two Windows features that genuinely reduce latency and improve gaming performance.

GAME MODE

Activates when a full-screen game is detected. Prioritizes the game process for CPU time, suppresses Windows Update installs during gameplay, and reduces background activity.

Check: **Settings** → **Gaming** → **Game Mode** → **On**. Or via PowerShell:

```
Set-ItemProperty -Path "HKCU:\Software\Microsoft\GameBar" -Name "AllowAutoGameMode" -Value 1
Set-ItemProperty -Path "HKCU:\Software\Microsoft\GameBar" -Name "AutoGameModeEnabled" -Value 1
```

HARDWARE ACCELERATED GPU SCHEDULING (HAGS)

Moves GPU memory scheduling off the CPU and onto the GPU directly — reducing CPU overhead and frame latency.

Requirements

REQUIREMENT	DETAILS
Windows version	Windows 10 version 2004 (May 2020 Update) or newer
NVIDIA GPU	RTX 2000 series (Turing) or newer + driver 451.48+
AMD GPU	RX 5000 series (RDNA 1) or newer + driver 20.5.1+

Enable via **admin PowerShell**:

```
Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\GraphicsDrivers" -Name "HwSchMode" -Value 2
```

Reboot after enabling HAGS. It will not take effect until Windows restarts.

// IF YOU SEE CRASHES OR BLACK SCREENS

Disable HAGS by setting the value back to 1 and rebooting:

```
Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\GraphicsDrivers" -Name "HwSchMode" -Value 1
```


Visual Effects & Animations

Windows wastes GPU and CPU on animations that serve no purpose while gaming.

METHOD 1: GUI

1. Right-click This PC → Properties → Advanced system settings
2. Under Performance click **Settings**
3. Select **Adjust for best performance**
4. Re-tick **Smooth edges of screen fonts** so text stays readable
5. Click Apply

METHOD 2: POWERSHELL

```
# Disable full window drag contents
Set-ItemProperty -Path "HKCU:\Control Panel\Desktop" -Name "DragFullWindows" -Value "0"
# Remove menu show delay
Set-ItemProperty -Path "HKCU:\Control Panel\Desktop" -Name "MenuShowDelay" -Value "0"
# Disable minimize/maximize animations
Set-ItemProperty -Path "HKCU:\Control Panel\Desktop\WindowMetrics" -Name "MinAnimate" -
Value "0"
# Disable taskbar animations
Set-ItemProperty -Path "HKCU:
\Software\Microsoft\Windows\CurrentVersion\Explorer\Advanced" -Name "TaskbarAnimations" -
Value 0
# Disable transparency/alpha selection in Explorer
Set-ItemProperty -Path "HKCU:
\Software\Microsoft\Windows\CurrentVersion\Explorer\Advanced" -Name "ListviewAlphaSelect"
-Value 0
# Disable desktop icon shadows
Set-ItemProperty -Path "HKCU:
\Software\Microsoft\Windows\CurrentVersion\Explorer\Advanced" -Name "ListviewShadow" -
Value 0
# Disable Aero Peek
Set-ItemProperty -Path "HKCU:\Software\Microsoft\Windows\DWM" -Name "EnableAeroPeek" -
Value 0
```

WINDOWS 11: TRANSPARENCY EFFECTS

Settings → Personalisation → Colours → Transparency effects → **Off**. This stops DWM continuously compositing transparent layers over your taskbar and Start menu.

Xbox Game DVR & Game Bar

Windows records your gameplay in the background. Turn it off if you do not use it.

WHAT IT DOES

Xbox Game DVR continuously buffers the last few minutes of gameplay footage in the background of every game you run. It is always encoding video, using CPU or GPU encoder resources, and writing to your drive — whether you asked it to or not.

DISABLE VIA SETTINGS

Settings → Gaming → Xbox Game Bar → Off

Settings → Gaming → Captures → Record in the background while I am playing a game → Off

DISABLE VIA REGISTRY

```
# Disable Game DVR
Set-ItemProperty -Path "HKCU:\System\GameConfigStore" -Name "GameDVR_Enabled" -Value 0

# Disable via policy (admin PowerShell)
If (!(Test-Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\GameDVR")) {
    New-Item -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\GameDVR" -Force
}
Set-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\GameDVR" -Name
"AllowGameDVR" -Value 0
```

// KEEP GAME MODE, DISABLE GAME DVR

These are separate settings. Game Mode (Ch.06) is beneficial — keep it on. Game DVR is recording — disable if unused.

Storage, SysMain & Temp Files

Keep drives clean, fast, and properly configured for your hardware type.

IDENTIFY YOUR DRIVE TYPE

```
Get-PhysicalDisk | Select-Object FriendlyName, MediaType, Size
```

SYSMAIN (SUPERFETCH)

Preloads frequently used apps into RAM ahead of time. Genuinely useful on **HDDs**. On **SSDs** it provides no benefit — drives are fast enough — but it still runs, causing constant background disk activity and RAM churn.

DRIVE TYPE	SYSMAIN
SSD (any type — SATA, NVMe)	Disable
HDD	Leave enabled
Mixed (SSD boot + HDD storage)	Disable

```
# Disable SysMain (admin PowerShell)
Stop-Service SysMain -Force
Set-Service SysMain -StartupType Disabled
```

TRIM FOR SSDS

TRIM keeps SSDs running at full speed over time by telling the drive which blocks are no longer needed. Verify it is enabled:

```
fsutil behavior query DisableDeleteNotify
# Result should be 0 (TRIM enabled). If 1, run:
fsutil behavior set DisableDeleteNotify 0
```

CLEAR TEMPORARY FILES

```
# Clear user temp folder
Remove-Item $env:TEMP\* -Recurse -Force -ErrorAction SilentlyContinue
```


Background Services

Windows runs dozens of services. Some are essential. Others you will never need.

// READ BEFORE PROCEEDING

Disabling the wrong service can cause instability or prevent Windows from booting. Only disable what you are certain you do not need. If unsure, set to Manual rather than Disabled.

HOW TO MANAGE SERVICES

Press Win + R → type `services.msc` → Enter. Double-click a service to change its startup type.

SAFE TO DISABLE ON MOST GAMING PCS

SERVICE	SAFE TO DISABLE IF...
Print Spooler	No printer connected
Fax	Always safe to disable
Remote Registry	Always safe on a home PC
Bluetooth Support Service	No Bluetooth devices used
Downloaded Maps Manager	Always safe to disable
Touch Keyboard and Handwriting	No touchscreen
Geolocation Service	Location features not needed

NEVER DISABLE THESE

SERVICE	WHY
Windows Defender / Security Center	Antivirus
Windows Update	Security patches
Windows Audio / Audio Endpoint Builder	Sound
DCOM Server / RPC	Core Windows communication

Display: Resolution, Refresh Rate & Sync

The most common mistake on a new install — never setting the correct refresh rate.

SET CORRECT RESOLUTION AND REFRESH RATE

Right-click desktop → Display settings → Advanced display settings:

- **Resolution:** Set to your monitor native (highest option)
- **Refresh rate:** Set to your monitor maximum (144Hz, 165Hz, 240Hz, etc)

// THIS IS MORE COMMON THAN YOU THINK

Many people buy 144Hz monitors and play at 60Hz because they never changed this setting.
Always check after a fresh install or GPU driver update.

G-SYNC (NVIDIA)

Open NVIDIA Control Panel → Display → Set up G-Sync. Tick Enable G-Sync / G-Sync Compatible → Enable for full screen mode → Apply.

In-game: turn VSync off. Cap frame rate 3–5fps below monitor refresh rate to stay inside the G-Sync range.

FREESYNC (AMD)

Open AMD Adrenalin → Display → AMD FreeSync → On. Same capping rule applies.

HDR

Settings → System → Display → Windows HD Color → Use HDR → On (if your monitor supports it).

Note: Windows desktop HDR can look washed out on some panels. Many prefer enabling it per-game only.

MULTIPLE MONITORS

Display settings are per-monitor. Select each monitor in Display settings and set correct resolution and refresh rate for each individually.

Mouse & Input Settings

Inconsistent mouse input destroys aiming accuracy. Fix it here.

DISABLE MOUSE ACCELERATION (POINTER PRECISION)

Mouse acceleration makes cursor travel distance depend on how fast you move the mouse — not just how far. For any game requiring consistent aim this is harmful. The same physical movement produces different outcomes depending on speed.

Control Panel → Hardware and Sound → Mouse → Pointer Options → uncheck Enhance pointer precision → Apply

SET POINTER SPEED TO 6/11

On the Pointer Options tab, set the speed slider to position **6 (middle)**. This is the only position where Windows applies no multiplier to raw mouse input. At any other position Windows is dividing or multiplying the hardware input. Adjust sensitivity in-game, not here.

ENABLE RAW INPUT IN GAMES

Raw Input is an option in many game settings menus. When enabled the game reads directly from the mouse hardware, bypassing all Windows mouse processing — no acceleration, no pointer speed, no filtering. **Always enable this where available.**

USB POLLING RATE

POLLING RATE	INTERVAL	NOTES
125 Hz	8ms	Outdated — upgrade if possible
500 Hz	2ms	Acceptable
1000 Hz	1ms	Recommended for gaming
4000 Hz+	0.25ms	High-end mice — marginal benefit

Set polling rate in your mouse software (Razer Synapse, Logitech G Hub, SteelSeries GG, etc.).

DPI

No universally correct DPI. Most competitive players use **400–3200 DPI** and adjust in-game sensitivity to taste. Consistency matters more than the specific number — pick one and stick to it.

GPU Driver Settings

The GPU control panel has significant performance and latency settings not optimized by default.

NVIDIA CONTROL PANEL

Right-click desktop → NVIDIA Control Panel → Manage 3D settings → Global Settings:

SETTING	VALUE	REASON
Power management mode	Prefer maximum performance	Prevents GPU downclocking mid-frame
Low Latency Mode	Ultra	Reduces pre-rendered frame queue, lowers input lag
Shader Cache Size	Unlimited	Caches compiled shaders, reduces stutter
Texture filtering — Quality	High performance	Faster texture sampling
Vertical sync	Off (control per-game)	Global VSync forces it everywhere including desktop
Background app max frame rate	20–30 fps	Saves GPU resources when tabbed out

// NVIDIA REFLEX

Available in many modern games. Enables it in-game alongside Low Latency Mode: Ultra in the control panel for the best latency reduction on NVIDIA hardware.

AMD ADRENALIN

Open AMD Adrenalin → Gaming → Global Graphics:

SETTING	VALUE	REASON
Radeon Anti-Lag	Enabled	Reduces CPU/GPU queue latency
Radeon Anti-Lag+	Enabled (where available)	Per-game tuned next-gen version
		Traditional VSync adds latency

Audio Optimization

Windows applies audio processing by default. Most of it adds latency and degrades positional accuracy.

DISABLE AUDIO ENHANCEMENTS

Windows applies equalizers, virtual surround, bass boost, and loudness equalization to all audio output by default. These add latency and distort directional cues.

1. Right-click speaker icon → **Sounds** → **Playback** tab
2. Double-click your active audio device
3. **Enhancements** tab → **Disable all enhancements** → **Apply**

Repeat for every audio device you use.

SET CORRECT SAMPLE RATE AND BIT DEPTH

In device properties → **Advanced** tab → set format to **24 bit, 48000 Hz (Studio Quality)**.

Mismatched sample rates force Windows to resample audio in real time, adding CPU overhead and introducing artifacts.

ENABLE EXCLUSIVE MODE

On the same Advanced tab, tick:

- **Allow applications to take exclusive control of this device**
- **Give exclusive mode applications priority**

This allows games to talk directly to the audio hardware, bypassing the Windows mixer — lower latency, better quality.

SPATIAL AUDIO

Windows Sonic, Dolby Atmos, and DTS:X simulate surround through stereo headphones. Whether useful depends on the game and personal preference. Enable per-device: right-click speaker icon → **Spatial sound**.

Network Optimizations

Lower ping, less jitter, and faster DNS — without changing hardware or ISP.

ETHERNET VS WIFI

The single biggest network improvement available to most people. A wired connection has lower and more consistent latency (1–2ms vs 5–30ms on WiFi), no interference, and no congestion from other wireless devices. A basic Cat5e or Cat6 cable is all you need. If running a cable is not possible, a **powerline adapter** is significantly more stable than WiFi for gaming.

CHANGE DNS SERVER

Network settings → Change adapter options → right-click adapter → Properties → IPv4 → Properties → Use the following DNS:

PROVIDER	PRIMARY	SECONDARY	NOTES
Cloudflare	1.1.1.1	1.0.0.1	Fastest globally, privacy-focused
Google	8.8.8.8	8.8.4.4	Very reliable, widely supported
Quad9	9.9.9.9	149.112.112.112	Blocks malicious domains

DISABLE NETWORK THROTTLING

Windows throttles network throughput for multimedia, which is counterproductive for gaming (admin PowerShell):

```
Set-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows  
NT\CurrentVersion\Multimedia\SystemProfile" -Name "NetworkThrottlingIndex" -Value  
0xffffffff
```

DISABLE WIFI POWER SAVING

If on WiFi, Windows may put the adapter into low-power mode causing periodic ping spikes: **Control Panel** → **Power Options** → **Change plan settings** → **Change advanced power settings** → **Wireless Adapter Settings** → **Power Saving Mode** → **Maximum Performance**.

Windows Defender & Security

Do not disable your antivirus. Configure it properly instead.

DO NOT DISABLE WINDOWS DEFENDER

Disabling Defender for "performance" is one of the most repeated pieces of bad advice in PC gaming communities. Modern Defender has a negligible impact on gaming frame rates, provides protection against real threats, and is far lighter than older versions.

The correct approach is to configure it properly, not disable it.

ADD GAME FOLDER EXCLUSIONS

Defender scans files as they are read and written. Adding game library folders as exclusions prevents scan-related hitches during map loads and shader compilation.

1. Windows Security → Virus & threat protection → Manage settings
2. Exclusions → Add or remove exclusions → Add an exclusion → Folder
3. Add each game library folder (e.g. D:\SteamLibrary, D:\Games)

// LIMIT EXCLUSIONS TO GAME FOLDERS ONLY

Do not exclude C:\ drive, Downloads folder, or Documents. Only dedicated game library directories.

SCHEDULE SCANS AT CONVENIENT TIMES

Open Task Scheduler → Task Scheduler Library → Microsoft → Windows → Windows Defender. Double-click the scheduled scan and set the trigger to a time when you are not gaming.

Monitoring Tools

You cannot optimize what you cannot measure.

MSI AFTERBURNER + RIVATUNER STATISTICS SERVER

The most widely used GPU monitoring and in-game overlay tool. Shows GPU usage, temperature, memory usage, frame rate, frame time, CPU usage, and RAM in real time. Free. RivaTuner Statistics Server is included — it handles the actual overlay rendering.

HWINFO64

Extremely detailed hardware monitoring. Every sensor on your system — per-core CPU temperatures, GPU hot spot temp, VRAM temperature, power draw, VRM temps. Use alongside Afterburner/RTSS for the most comprehensive overlay. Free.

LATENCYMON

Measures system DPC (Deferred Procedure Call) latency and identifies which drivers are causing it. High DPC latency causes audio stutters and frame time spikes. If you have unexplained stutters, run LatencyMon for a few minutes — it identifies the exact driver culprit. Free.

GPU-Z AND CPU-Z

Read-only hardware information. GPU-Z shows exact GPU model, VRAM, clocks, and sensor data. CPU-Z shows CPU details and memory speed — useful for verifying XMP is working. Both free.

WHAT TO WATCH WHILE GAMING

METRIC	CONCERN IF...
GPU Usage	Below 95% with low FPS — CPU bottleneck likely
GPU Temperature	Above 85°C — improve airflow or repaste
Frame Time	Spikes cause stutters even if average FPS is high
VRAM Usage	Hitting the cap causes severe stutters — lower texture settings
RAM Usage	Near capacity consistently — consider adding RAM

Ongoing Maintenance

A well-optimized PC stays optimized with regular maintenance.

WEEKLY

- Check for Windows updates
- Check for GPU driver updates — major releases often improve game performance significantly

MONTHLY

- Clear temp files
- Run Disk Cleanup (search in Start → also clean up system files)
- Review startup programs — newly installed software often adds itself without asking
- Check installed programs for anything unused or unrecognized

EVERY 6–12 MONTHS

- Clean dust from PC internals and fans — dust causes thermal throttling
- Check drive health: `Get-PhysicalDisk | Select-Object FriendlyName, HealthStatus`
- Remove programs not opened in 6+ months

SIGNS YOUR PC NEEDS ATTENTION

SYMPTOM	LIKELY CAUSE	ACTION
Progressively slower boot	Accumulated startup programs	Audit startup list
Sudden FPS drops	Thermal throttling or background process	Check temps and Task Manager
Audio crackling	High DPC latency	Run LatencyMon
Random BSODs	Driver issue, overheating, or failing hardware	Check Event Viewer, run memtest
High ping spikes	Network driver or background download	Check Task Manager network tab

Final Checklist

Work through this page. Every tick is an improvement.

SETUP & FOUNDATION

TASK	□
Privacy settings configured on first boot	☐
Windows fully updated — no pending updates	☐
Chipset drivers installed from manufacturer site	☐
GPU drivers installed with clean install	☐
Audio drivers from motherboard support page	☐
XMP / EXPO enabled in BIOS and verified	☐
Resizable BAR enabled (if supported)	☐

WINDOWS TWEAKS

TASK	□
Power plan set to High or Ultimate Performance	☐
Startup programs audited and trimmed	☐
Game Mode enabled	☐
HAGS enabled and PC rebooted	☐
Visual effects set to Best Performance	☐
Xbox Game DVR disabled	☐
SysMain disabled (SSD users)	☐
TRIM enabled for SSDs	☐

